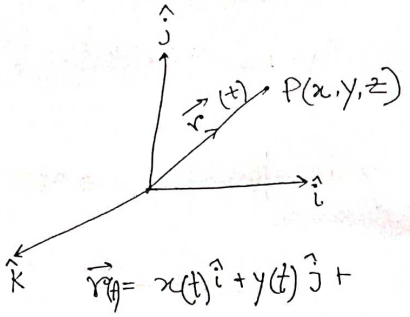
24/6/2024
Monday

Kinematics

Position



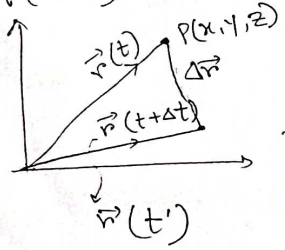
$\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$

In 3D:

$\vec{r}(t)$

In 1D: $\vec{x}(t)$

Displacement (সরণ)



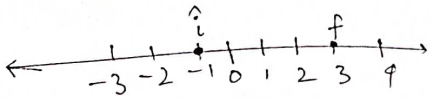
$\boxed{\Delta t = t' - t}$

$\Delta\vec{r} = \vec{r}(t+\Delta t) - \vec{r}(t)$

$\vec{r}(t) + \Delta\vec{r} = \vec{r}(t+\Delta t)$

$\therefore \Delta\vec{r} = \vec{r}(t+\Delta t) - \vec{r}(t)$

$= \vec{r}(t_f) - \vec{r}(t_i)$



$\Delta\vec{r} = 3\hat{i} - (-\hat{i}) = 4\hat{i}$

Total distance travelled

$\Delta s \rightarrow$ actual scalar distance travelled

Velocity

Average velocity

$$\vec{V}_{avg} = \frac{\Delta \vec{r}}{\Delta t}$$

Average speed

$$V_{avg} = \frac{\Delta s}{\Delta t}$$

$$\frac{d}{dx} f(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Instantaneous Velocity

$$\vec{V} = \lim_{\Delta t \to 0} \vec{V}_{avg} = \lim_{\Delta t \to 0} \frac{\Delta \vec{r}}{\Delta t}$$

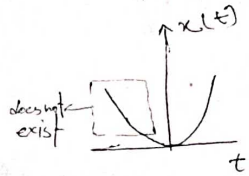
$$= \lim_{\Delta t \to 0} \frac{\vec{r}(t+\Delta t) - \vec{r}(t)}{\Delta t}$$

$$= \frac{d\vec{r}}{dt}$$

$$\boxed{\vec{V}(t) = \frac{d\vec{r}(t)}{dt}}$$

1D $\vec{x}(t) = t^2 \hat{i}$

Find the velocity of the particle at $t = 3s$.



$$\vec{V}_x(t) = \frac{d\vec{x}(t)}{dt}$$

$$\vec{V}_x(t) = \frac{d}{dt}(t^2)\hat{i} = 2t\hat{i}$$

$$\boxed{\therefore \vec{V}_x(3) = 6\hat{i}}$$

Instantaneous speed, $v = \frac{ds}{dt}$

3D

$$\vec{r}(t) = 2t\,\hat{i} - t^2\hat{j} + 3\hat{k}$$

Find ~~veco~~ velocity

$$\frac{d\vec{r}(t)}{dt} = 2\hat{i} - 2t\hat{j}$$

$$\cancel{\sqrt{2^2 + 1^2} = \sqrt{9} = \sqrt{3}}$$

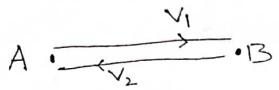
$$\vec{V}(t) = \frac{d\vec{r}(t)}{dt}$$

$$\vec{V}(t) = 2\hat{i} - 2t\hat{j} + 0\hat{k}$$

$$\boxed{\vec{V}(t) = V_x\hat{i} + V_y\hat{j} + V_z\hat{k}}$$

$$V_x = \frac{dx(t)}{dt}, \quad V_y = \frac{dy(t)}{dt}, \quad V_z = \frac{dz(t)}{dt}$$

#



For constant speed
$d = vt$

(i) Displacement $= \vec{0}$

(ii) Total distance travelled $= \cancel{V_1 + V_2}$ $\cancel{AB+BA}$ $2AB$

(iii) Average velocity $= \vec{0}$

(iv) Average speed (in term of V_1 and V_2)

$$\text{Average speed} = \frac{\Delta s}{\Delta t} = \frac{2AB}{\Delta t} = \frac{2AB}{t_1 + t_2}$$

$AB = V_1 t_1$

$BA = V_2 t_2$

$\therefore t_1 = AB/V_1$

$t_2 = AB/V_2$

$$= \frac{2AB}{\frac{AB}{V_1} + \frac{AB}{V_2}} = \boxed{\frac{2V_1V_2}{V_1 + V_2}}$$

$* \rightarrow 1$

$\# \rightarrow 1$

$* \rightarrow 100$

$\# \rightarrow 20$

120

Average $= \frac{120}{120}$

$= 1$

$* \rightarrow 5$

$\# \rightarrow 0.5$

500

10

510

Weighted Average $\frac{510}{120} = 4.25$

Acceleration

Average acceleration, $\boxed{\vec{a}_{avg} = \frac{\Delta \vec{v}}{\Delta t}}$

Instantaneous acceleration $\boxed{\vec{a} = \frac{d\vec{v}}{dt}}$

$$\vec{a} = a_x \hat{i} + a_y \hat{j} + a_z \hat{k} = \frac{d}{dt}\left(\frac{d\vec{r}}{dt}\right)$$

$$a_x = \frac{dv_x}{dt}, \; a_y = \frac{dv_y}{dt}, \; a_z = \frac{dv_z}{dt} = \frac{d^2\vec{r}}{dt^2}$$

※

$$a(t) = \frac{d\vec{v}(t)}{dt}$$

$$\Rightarrow \int_{t_0}^{t} \vec{a}(t)\,dt = \int_{\vec{v}(t_0)}^{\vec{v}(t)} \frac{d\vec{v}(t)}{dt}\,dt$$

$$\Rightarrow \int_{t_0}^{t} \vec{a}(t)\,dt = \vec{v}(t)\Big|_{\vec{v}(t_0)}^{\vec{v}(t)}$$

$$= \int_{t_0}^{t} \vec{a}(t)\,dt = \vec{v}(t) - \vec{v}(t_0)$$

$$\boxed{\therefore \vec{v}(t) = \vec{v}(t_0) + \int_{t_0}^{t} \vec{a}(t)\,dt}$$

Special cases

1. If $t_0 = 0$, $\vec{v}(t) = \vec{v}(0) + \int_0^t \vec{a}(t)\,dt$

$$\boxed{\therefore\ \vec{v}(t) = \vec{v}_0 + \int_0^t \vec{a}(t)\,dt}$$

2. If $\vec{a}(t)$ = constant

$$\boxed{\vec{v}(t) = \vec{v}_0 + at}$$

$$\# \quad \vec{v}(t) = \frac{d\vec{r}(t)}{dt}$$

$$\Rightarrow \int_0^t \vec{v}(t)\,dt = \int_{\vec{r}_0}^{\vec{r}(t)} \frac{d\vec{r}(t)}{dt}\,dt$$

$$\boxed{\therefore\ \vec{r}(t) = \vec{r}_0 + \int_0^t \vec{v}(t)\,dt}$$

Special Case:

1. If $\vec{a}(t)$ = constant,

$$\vec{r}(t) = \vec{r}_0 + \int_0^t (\vec{v}_0 + \vec{a}t)\,dt$$

$$\Rightarrow \vec{r}(t) = \vec{r}_0 + \int_0^t \vec{v}_0\,dt + \int_0^t \vec{a}\,t\,dt$$

H.W

$$\boxed{\therefore\ \vec{r}(t) = \vec{r}_0 + \vec{v}_0 t + \frac{1}{2}\vec{a}\,t^2}$$

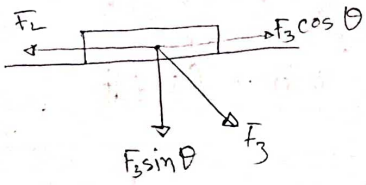
Quiz 1 Monday 01/07/2024

Everything taught till today

Tutorial

21/6/21 Monday

libretexts



$F_3\cos\theta - F_2 = ma_x$

Wednesday

A particle moves under a time dependent acceleration $\vec{a}(t) = 3t\hat{i}$. It's position at time $t=0$ was $3m\hat{i} - 2m\hat{j}$ and ~~vecoti~~ velocity was $-5ms^{-1}\hat{i} + 2ms^{-1}\hat{j}$

(i) Find the velocity of the particle at $t=2s$

(ii) " " position " " " " $t=3s$

$$\vec{v}(t) = \vec{v}_0 + \int_0^t \vec{a}(t)\,dt$$

$$\vec{r}(t) = \vec{r}_0 + \int_0^t \vec{v}(t)\,dt$$

Soln:

(i) $\vec{v}_0 = -5\hat{i} + 2\hat{j}$

$$\vec{v}_{(2)} = (-5\hat{i} + 2\hat{j}) + \int_0^2 (3t\hat{i})\,dt$$

$$= -5i + 2j + \int_0^2 3t\,dt\,\hat{i}$$

$$\boxed{\vec{v}(t) = -5\hat{i} + 2\hat{j} + 3\frac{t^2}{2}\hat{i}}$$

$$\therefore \boxed{\vec{v}_{(2)} = 1ms^{-1}\hat{i} + 2ms^{-1}\hat{j}}$$

(11) $\vec{r}_0 = 3\hat{i} - 2\hat{j}$

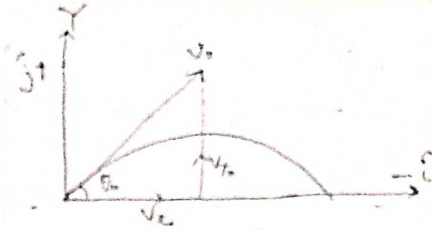
$$\vec{r}(3) = 3\hat{i} - 2\hat{j} + \int_0^t \left(-5\hat{i} + 2\hat{j} + \frac{3t^2}{2}\hat{i}\right) dt$$

$$\vec{r}(t) = (3\hat{i} - 2\hat{j}) - 5t\hat{i} + 2t\hat{j} + \frac{3t^3}{6}\hat{i}$$

$$\therefore \vec{r}(3) = 3\hat{i} - 2\hat{j} - 5\times 3\hat{i} + 2\times 3\hat{j} + \frac{3(3)^3}{6}\hat{i}$$

$$= 3\hat{i} - 2\hat{j} - 15\hat{i} + 6\hat{j} + \frac{81}{6}\hat{i}$$

Projectile motion



$$\vec{r}_0 = x_0\hat{i} + y_0\hat{j}$$

$$\vec{v}_0 = v_{x_0}\hat{i} + v_{y_0}\hat{j}$$

$$\boxed{\therefore \vec{v}_0 = v_0\cos\theta_0\,\hat{i} + v_0\sin\theta_0\,\hat{j}}$$

$$\vec{a}(t) = 0\hat{i} - g\hat{j}$$

$$\vec{v}(t) = \vec{v}_0 + \vec{a}t = (v_0\cos\theta_0\,\hat{i} + v_0\sin\theta_0\,\hat{j}) + (0\hat{i} - g\hat{j})t$$

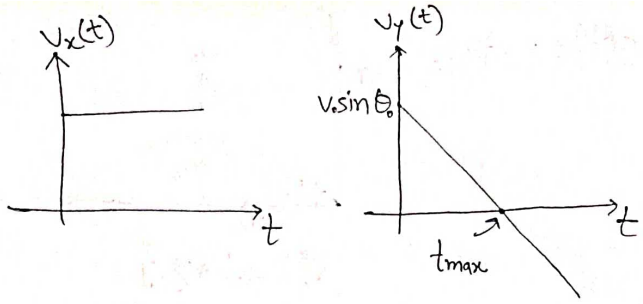
$$\boxed{\therefore \vec{v}(t) = v_0\cos\theta_0\,\hat{i} + (v_0\sin\theta_0 - gt)\hat{j}}$$

$$\vec{v}(t) = v_x(t)\hat{i} + v_y(t)\hat{j}$$

$$\therefore v_x(t) = v_0\cos\theta_0$$

$$v_y(t) = v_0\sin\theta_0 - gt$$

$$Y = c - mx$$



$$V_y(t) = V_0 \sin\theta_0 - gt$$

$$\Rightarrow V_y(t_{max}) = V_0 \sin\theta_0 - g t_{max}$$

$$\Rightarrow 0 = V_0 \sin\theta_0 - g t_{max}$$

$$\boxed{\therefore t_{max} = \frac{V_0 \sin\theta_0}{g}}$$

Projectile motion

$$\vec{r}(t) = \vec{r}_0 + \vec{v}_0 t + \frac{1}{2}\vec{a}t^2$$

$$= (x_0\hat{i} + y_0\hat{j}) + (V_0\cos\theta_0 t\hat{i} + V_0\sin\theta_0 t\hat{j}) + \frac{1}{2}(0\hat{i} - g\hat{j})t^2$$

$$\boxed{\therefore \vec{r}(t) = (x_0 + V_0\cos\theta_0 t)\hat{i} + (y_0 + V_0\sin\theta_0 t - \tfrac{1}{2}gt^2)\hat{j}}$$

(iii) $\boxed{x(t) = x_0 + V_0\cos\theta_0 t}$

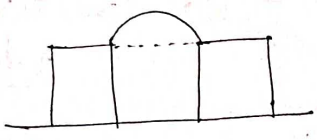
(iv) $\boxed{y(t) = y_0 + V_0\sin\theta_0 t - \frac{1}{2}gt^2}$

(i) $V_x(t) = V_0\cos\theta_0$ (ii) $V_y(t) = V_0\sin\theta_0 - gt$

Flight time (if the particle lands at same height)

$T = 2t_{max}$

$$\boxed{\therefore T = \frac{2v_0 \sin\theta_0}{g}}$$



Maximum height from ~~the~~ launching point:

$$y(t) = 0 + v_0 \sin\theta_0 t - \frac{1}{2} g t^2$$

$$\Rightarrow y(t_{max}) = v_0 \sin\theta_0 t_{max} - \frac{1}{2} g t_{max}^2$$

$$\Rightarrow y_{max} = v_0 \sin\theta_0 \frac{v_0 \sin\theta_0}{g} - \frac{1}{2} \frac{v_0^2 \sin^2\theta_0}{g^2}$$

Range (if the particle lands at the same height)

Maximum horizontal distance travelled.

$$x(t) = 0 + v_0 \cos\theta_0 t$$

$$\Rightarrow x(T) = v_0 \cos\theta_0 T$$

$$\Rightarrow R = \frac{v_0 \cos\theta_0 \times 2 v_0 \sin\theta_0}{g}$$

$$\Rightarrow R = \frac{v_0^2 \, 2 \sin\theta_0 \cos\theta_0}{g}$$

$$\boxed{\therefore R = \frac{v_0^2 \sin 2\theta_0}{g}}$$

$R = R_{max}$, if $\sin 2\theta_0 = max$

$\Rightarrow \sin 2\theta_0 = 1$

$\Rightarrow \sin 2\theta_0 = \sin 90^\circ$

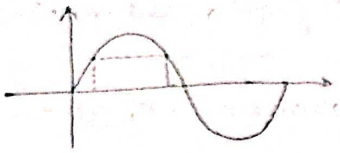
$$\boxed{\therefore \theta = 45^\circ}$$

* $\sin(180^\circ - \theta) = \sin\theta$

$\therefore \sin(180^\circ - 2\theta_0) = \sin 2\theta_0$

$2\theta_0' = 180^\circ - 2\theta_0$

$\boxed{\therefore \theta_0' = 90^\circ - \theta_0}$



$\begin{cases} \theta_0 = 30^\circ \longrightarrow R \\ \theta_0 = 90^\circ - 30^\circ = 60^\circ \longrightarrow R \end{cases}$

$\begin{cases} \theta_0 = 10^\circ \longrightarrow R \\ \theta_0 = 80^\circ \longrightarrow R \end{cases}$

Particles thrown directly upward

$\theta_0 = 90^\circ$

(i) $x(t) = x_0$ (ii) $y(t) = y_0 + v_0 t - \frac{1}{2} g t^2$

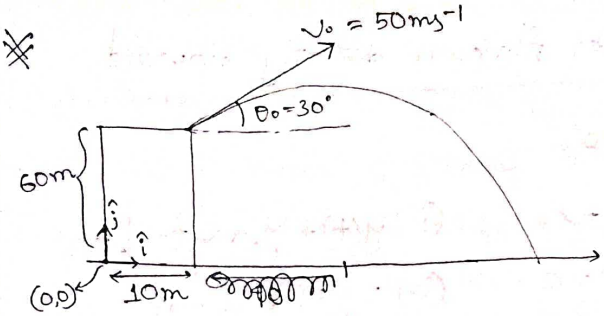
(iii) $v_x(t) = 0$ (iv) $v_y(t) = v_0 - gt$

* $t_{max} = \frac{v_0 \sin\theta_0}{g} = \frac{v_0 \sin 90^\circ}{g}$

$\boxed{\therefore t_{max} = \frac{v_0}{g}}$

* $T = \frac{2v_0}{g}$ * $y_{max} = \frac{v_0^2}{2g}$ * $R = 0$

※



(i) Find the velocity and position at $t = 2s$ after the launch.

(ii) Find the time of flight

(iii) Find the horizontal distance travelled

Solution: $\vec{V}(t) = \ldots$

(H.W.) $\vec{r}(t) = \ldots$

$x_0 = 10$

$y_0 = 60$

(ii) $y_0 = 60$ $\quad y(T) = 0$

$\therefore y(t) = y_0 + v_0 \sin\theta_0 t - \frac{1}{2} g t^2$

$\Rightarrow y(T) = y_0 + v_0 \sin\theta_0 T - \frac{1}{2} g t^2$

$\Rightarrow 0 = 60 + 25T - 4.9T^2$

$$\boxed{\therefore 4.9T^2 - 25T - 60 = 0}$$

$$T = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

(iii) $x(t) = x_0 + v_0 \cos\theta_0 t$

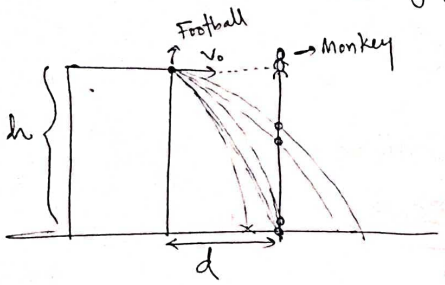
$\therefore x(T) = 0 + v_0 \cos\theta_0 T$

$$\boxed{\therefore x(T) = v_0 \cos\theta_0 T}$$

1/7/24 ~~Wednesday~~

Shoot the monkey

At what speed the football has to be shot such that it hits the monkey?



$x_0 = 0$ | $x = d$

$y_0 = 0$ | $y = -h$

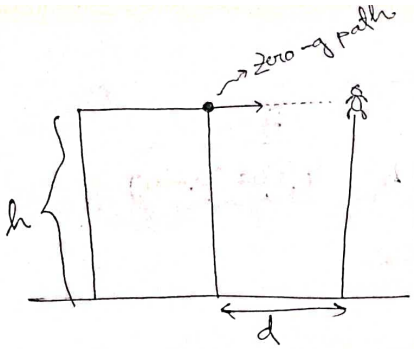
$$\vec{x}(t) = \vec{x}_0 + v_0 \cos\theta_0 t\,\hat{i}$$

$$\Rightarrow d\hat{i} = 0\hat{i} + v_{0,min} \cos 0^\circ t\,\hat{i}$$

$$\boxed{\therefore v_{0,min} = \frac{d}{t}} = 4.042\,ms^{-1}$$

$$\vec{y}(t) = \vec{y}_0 + (v_0 \sin\theta_0 t - \tfrac{1}{2}gt^2)\hat{j}$$

$$\Rightarrow -h\hat{j} = 0\hat{j} + (0 - \tfrac{1}{2}gt^2)\hat{j} \qquad \boxed{\therefore t = \sqrt{\frac{2h}{g}}} = 2.474\,s$$



Always hit the monkey if $v_0 \geq v_{0,min}$

Zero-g

Football: y_0

$\vec{y}(t) = \vec{y}_0$

Monkey: y_0

$\boxed{\vec{y}(t) = \vec{y}_0 - \frac{1}{2}gt^2}$

$\vec{y}(t) = \vec{y}_0$

With-g

Football: y_0

$\vec{y}(t) = \vec{y}_0 - \frac{1}{2}gt^2\,\hat{j}$

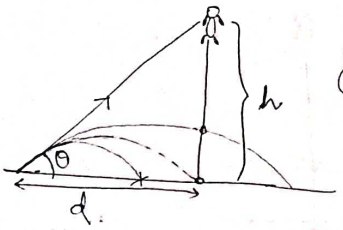
$\boxed{\Delta y = \frac{1}{2}gt^2}$

Monkey: y_0

$\vec{y}(t) = \vec{y}_0 - \frac{1}{2}gt^2\,\hat{j}$

$\boxed{\therefore \Delta y = \frac{1}{2}gt^2}$

#



H.W.
Find $V_{0,min}$

2. $r = \sqrt{3^2+4^2} = 5$ $\quad \theta = \tan^{-1}\left(\frac{4}{3}\right) = 53 \cdots$

$\theta = 180° - 53 \cdots$

3. 100 units

4. $\vec{v} = \frac{d\vec{r}}{dt} = \frac{d}{dt}(5t\hat{i} - t^2\hat{j}) = 5\hat{i} - 2t\hat{j}$

$\vec{a} = \frac{d\vec{v}}{dt} = 0\hat{i} - 2\hat{j}$

5. $2\vec{A} \cdot 4\vec{B} = 8(\vec{A} \cdot \vec{B}) = 8 \times 20 = 160$

6. (a) $(\vec{A} \cdot \vec{A}) + \vec{B}$ $\quad$ (b) $\vec{A} \times (\vec{A} \cdot \vec{B})$

3/7/24
Wednesday

Newtonian Mechanics

↑

Classical Mechanics $\xrightarrow{a \approx 0}$ Quantum Mechani

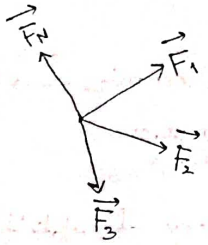
↓ c = max. speed $= 3 \times 10^8\ ms^{-1}$

Special Theory of Relativity

Force

- → A vector quantity
- → Tries to change the velocity of an object
- → Occurs in pair

Superposition principle



$$\vec{F}_{net} = \sum_{i=1}^{N} \vec{F}_i$$

$$= \vec{F}_1 + \vec{F}_2 + \dots + \vec{F}_N$$

Newton's Laws

First Law: If there is no net external force applied on an object, the velocity of the object remains constant.

$$\sum \vec{F} = \vec{0} \ , \ \vec{v} = \text{constant}$$

Second Law:

Momentum $\vec{P} = m\vec{v}$

$$\frac{d\vec{P}}{dt} \propto \vec{F}$$

$$\Rightarrow \frac{d\vec{P}}{dt} = k\vec{F}$$

$$\boxed{\therefore \vec{F} = \frac{d\vec{P}}{dt}}$$

$$\vec{F} = \frac{d}{dt}(m\vec{v})$$

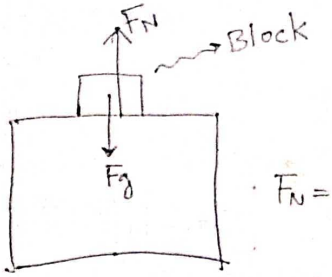
$$\Rightarrow \vec{F} = m\frac{d\vec{v}}{dt} + \vec{v}\frac{dm}{dt}$$

If m = constant,

$$\boxed{\therefore \vec{F} = m\vec{a}} \quad \boxed{\Sigma\vec{F} = m\vec{a}}$$

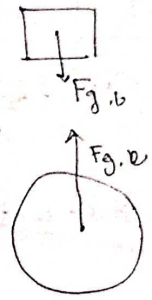
<u>Third Law:</u>

Every action force has a reaction force associated with it, but the <u>action and reaction force acts on different objects.</u>

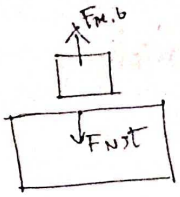


F_N = normal reaction force

Are F_g and F_N Newton's third law pair?



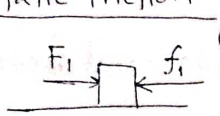
$$\boxed{F_{g.b} = F_{g.e}}$$



$$\boxed{F_{N.b} = F_{N.t}}$$

Friction

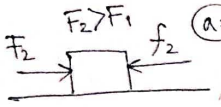
Static Friction



$a = 0$

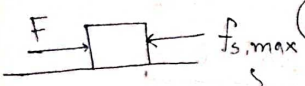
$\boxed{\therefore f_{1,s} = F_1}$

$f \rightarrow$ Static Friction force



$a =$

$\boxed{\therefore f_{2,s} = F_2}$



$a = 0 \rightarrow$ About to move

maximum static Friction force

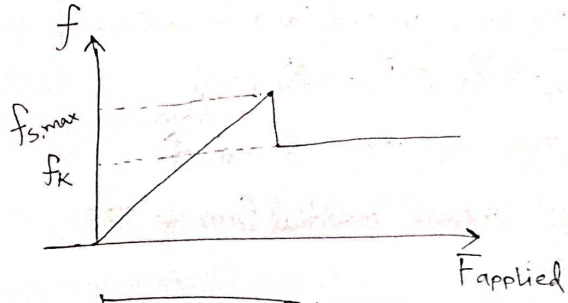
$$\boxed{f_{s,max} = \mu_s F_N}$$

μ_s = co-efficient of static friction

Kinetic friction force.

$$\boxed{f_K = \mu_K F_N}$$

μ_K = co-efficient of Kinetic friction



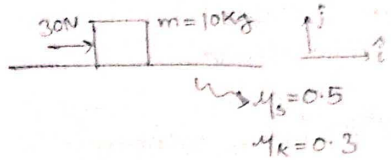
$\boxed{\mu_K < \mu_s} \rightarrow \boxed{f_K < f_{s,max}}$

Procedures for solving force-related problems

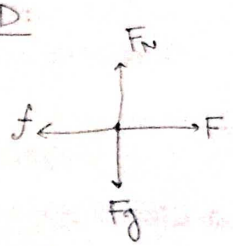
1. Set up prefered coordinate system.
2. Draw the free body diagram.
3. Divide the forces along coordinate axes.
4. Write Newtons second law for indivisual axes ($\Sigma \vec{F}_x = m\vec{a}_x$, $\Sigma \vec{F}_y = m\vec{a}_y$, etc)
5. Solve the equation found from (4)
6. Always choose inertial frame of reference.

Example 1

Find the acceleration of the block



FBD:



$\hat{j}: \Sigma \vec{F}_y = m\vec{a}_y$

$\Rightarrow F_N \hat{j} + F_g(-\hat{j}) = m \times 0\hat{j}$

$F_N = F_g$ $\quad \boxed{F_N = mg = 98\,N}$

$f_{s,max} = \mu_s F_N = 49\,N$

$f_k = \mu_k F_N = 29.4\,N$

Condition for acceleration:

$$\boxed{F_{app} > f_{s,max}}$$

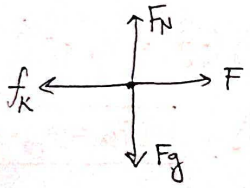
Checking condition : $\boxed{F_{app} < f_{s,max}}$

Not satisfied

$$\boxed{\therefore a = 0}$$

If $F = 60N$, then $a = ?$

$\boxed{F_{app} > f_{s,max}}$ → Satisfied



$$\Sigma \vec{F}_x = m\vec{a}$$

$$\hat{i} : F\hat{i} - f_K\hat{i} = ma_x\hat{i}$$

$$\therefore a_x = \frac{F - f_K}{m} = \frac{60 - 27.4}{10} \, ms^{-2}$$

$$= 3.06 ms^{-2}$$